電気回路Ⅲ　　解答

23.1

(1) 位相差 $\theta = \theta_V - \theta_I = -30 - 30 = -60°$

$P = VI\cos\theta = 100 \times 2 \times \dfrac{1}{2} = 100$ [W]

$P_r = VI\sin\theta = 100 \times 2 \times \left(-\dfrac{\sqrt{3}}{2}\right) = -100\sqrt{3}$

$\fallingdotseq -173$ [var]　$P_a = VI = 200$ [VA]

(2) $\dot{P} = \dot{V}\bar{\dot{I}} = (2+j4)(2-j1) = 8+j6$

$P = \mathrm{Re}(\dot{P}) = 8$ [W]　$P_r = \mathrm{Im}(\dot{P}) = 6$ [var]

$P_a = |\dot{P}| = \sqrt{8^2+6^2} = 10$ [VA]

(3) $Z = |\dot{Z}| = \sqrt{10}$ [Ω]　$I = \dfrac{V}{Z} = \sqrt{10}$ [A]

$P = RI^2 = 30$ [W]　$P_r = XI^2 = 10$ [var]

$P_a = VI = 10\sqrt{10} \fallingdotseq 31.6$ [VA]

(4) $P_a = \sqrt{P^2+P_r^2} = 20$ [VA]　$I = \dfrac{P_a}{V} = 2$ [A]

$P = P_a\cos\theta$ より $\cos\theta = \dfrac{P}{P_a} = \dfrac{16}{20} = 0.8$

(5) $P = VI\cos\theta$ より $V = \dfrac{P}{I\cos\theta} = \dfrac{60}{2\times 0.6} = 50$ [V]

$Z = \dfrac{V}{I} = \dfrac{50}{2} = 25$ [Ω]　$\dot{Z}$ は容量性なので

$\theta < 0$ だから，$\sin\theta = -\sqrt{1-\cos^2\theta} = -0.8$

$\dot{Z} = Z\angle\theta = Z(\cos\theta + j\sin\theta) = 15 - j20$ [Ω]

23.2

(1) $\dot{Z} = R + j\omega L = 6 + j8$ [Ω]　$Z = |\dot{Z}| = 10$ [Ω]

$I = \dfrac{E}{Z} = 2$ [A]　$P = RI^2 = 6\times 4 = 24$ [W]

$P_r = XI^2 = 8\times 4 = 32$ [var]　$P_a = EI = 40$ [VA]

(2) $\dot{I} = \left(\dfrac{1}{R} + j\omega C\right)\dot{E} = (0.1 + j0.2)\times 10 = 1 + j2$ [A]

$\dot{P} = \dot{E}\bar{\dot{I}} = 10(1-j2) = 10 - j20$　$\therefore P = 10$ [W]

$P_r = -20$ [var]　$P_a = |\dot{P}| = 10\sqrt{5} \fallingdotseq 22.4$ [VA]

(3) RL の電流 $\dot{I}_{RL} = \dfrac{\dot{E}}{R + j\omega L} = \dfrac{10}{1+j3} = 1 - j3$

C の電流 $\dot{I}_C = j\omega C\dot{E} = j0.2 \times 10 = j2$

$\dot{I} = \dot{I}_{RL} + \dot{I}_C = 1 - j1$ [A]　$\dot{P} = \dot{E}\bar{\dot{I}} = 10 + j10$

$\therefore P = 10$ [W]　$P_r = 10$ [var]　$P_a = 10\sqrt{2}$

$\fallingdotseq 14.1$ [VA]

23.3　$\dot{Z}$ は誘導性なので，偏角 $\theta > 0$

$\cos\theta = 0.8 \rightarrow \sin\theta = \sqrt{1-\cos^2\theta} = 0.6$

$P = P_a\cos\theta$ より，$P_a = \dfrac{P}{\cos\theta} = \dfrac{200}{0.8} = 250$ [VA]

$P_r = P_a\sin\theta = 250 \times 0.6 = 150$ [var]　$I = \dfrac{P_a}{V} = \dfrac{250}{50}$

$= 5$ [A]　$Z = \dfrac{V}{I} = \dfrac{50}{5} = 10$ [Ω]　$\dot{Z} = Z\angle\theta$

$= Z(\cos\theta + j\sin\theta) = 10(0.8 + j0.6) = 8 + j6$ [Ω]